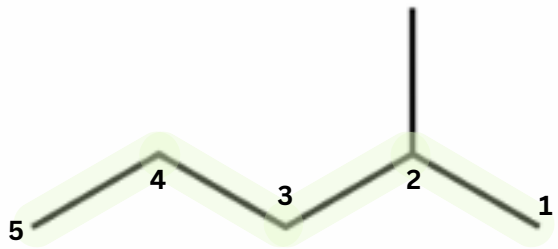
A01

STRUCTURE → NAME

Level: Foundation

Molecular Formula: C_6H_{14}

1. NUMBERED PARENT



Parent: 5-carbon chain highlighted · numbered from the right end.

2. NAMING TRAIL

Parent:

Longest chain = 5 C → pentane.

Numbering:

Right to left: methyl on C2. Left to right: methyl on C4. First point of difference: 2 is lower than 4, so number from the right.

Substituents:

Methyl on C2 → 2-methyl.

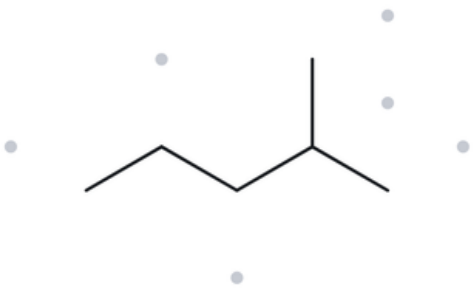
3. MY NAME (WRITTEN FIRST)

2-methylpentane

4. THE ALKANE CHECK

Returned name:

2-methylpentane



✓ Match

Verify ☑ longest chain/ring ☑ aphabetical order ☑ formula counted ☑ lowest locants ☑ punctuation

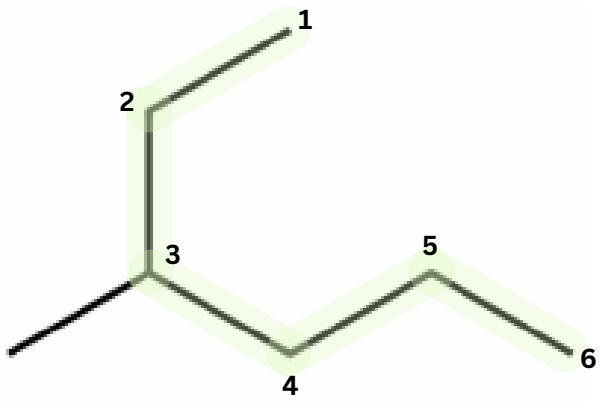
A02

STRUCTURE → NAME

Level: Foundation

Molecular Formula: C_7H_{16}

1. NUMBERED PARENT



Parent: 6-carbon chain highlighted · numbered from the upper end.

2. NAMING TRAIL

Parent:

Longest continuous chain = 6 C → hexane.

Numbering:

From the upper end: methyl on C3. From the opposite end: methyl on C4. First point of difference: 3 is lower than 4, so number from the upper end.

Substituents:

Methyl on C3 → 3-methyl.

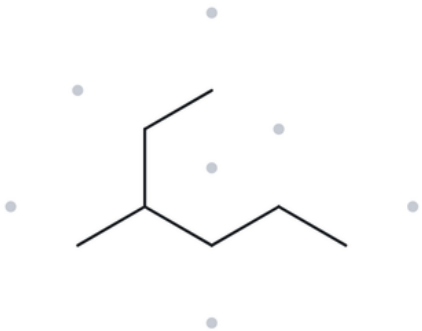
3. MY NAME (WRITTEN FIRST)

3-methylhexane

4. THE ALKANE CHECK

Returned name:

3-methylhexane



✓ Match

Verify ☑ longest chain/ring ☑ aphabetical order ☑ formula counted ☑ lowest locants ☑ punctuation

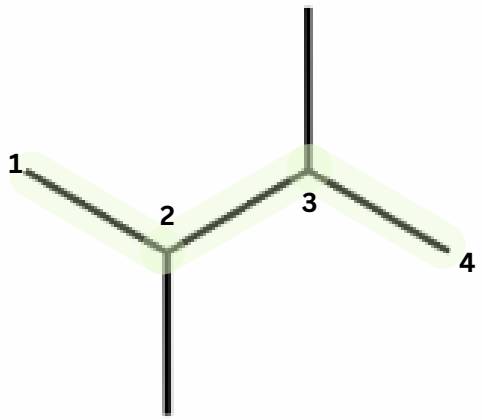
A03

STRUCTURE → NAME

Level: Foundation

Molecular Formula: C_6H_{14}

1. NUMBERED PARENT



Parent: 4-carbon chain highlighted · numbered from either end.

2. NAMING TRAIL

Parent:

Longest chain = 4 C → butane.

Numbering:

Left to right: methyl groups on C2 and C3.
Right to left: methyl groups on C2 and C3.
Both directions give the same locants.

Substituents:

Two methyl groups on C2 and C3 → 2,3-dimethyl.

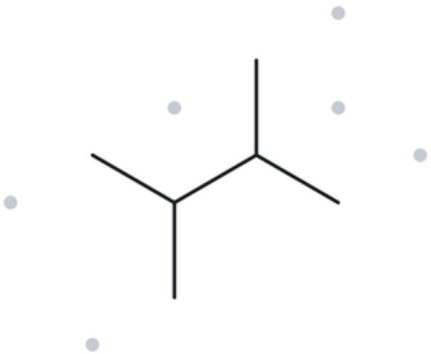
3. MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4. THE ALKANE CHECK

Returned name:

2,3-dimethylbutane



✓ Match

Verify

☑ longest chain/ring ☑ aphabetical order ☑ formula counted
☑ lowest locants ☑ punctuation

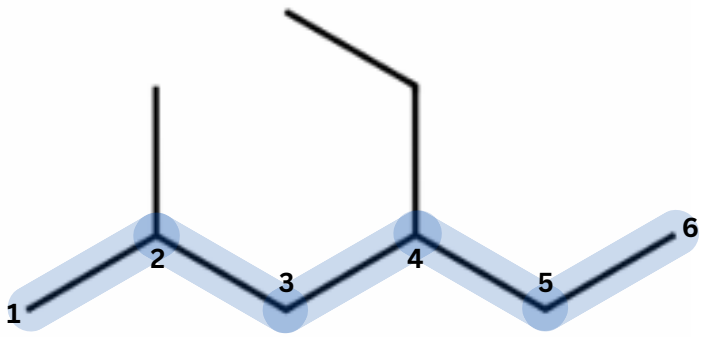
A04

STRUCTURE → NAME

Level: Developing

Molecular Formula: C_9H_{20}

1. NUMBERED PARENT



Parent: 6-carbon chain highlighted · numbered from the end nearer the substituents.

2. NAMING TRAIL

Parent:

Longest continuous chain = 6 C → hexane.

Numbering:

Numbering from the methyl-substituted end gives locants 2 and 4. Numbering from the opposite end gives locants 3 and 5. The set 2,4 is lower.

Substituents:

Ethyl on C4; methyl on C2. Alphabetical order: ethyl before methyl.

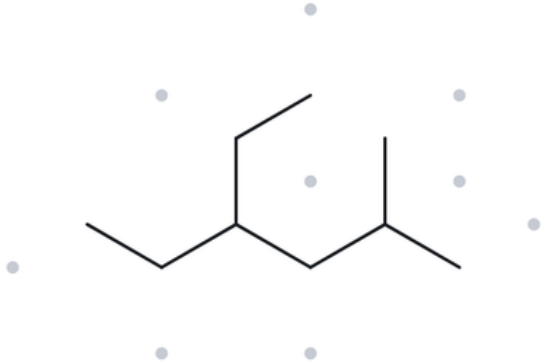
3. MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexaneylpentane

4. THE ALKANE CHECK

Returned name:

4-ethyl-2-methylhexane



✓ Match

Verify ☑ longest chain/ring ☑ aphabetical order ☑ formula counted ☑ lowest locants ☑ punctuation

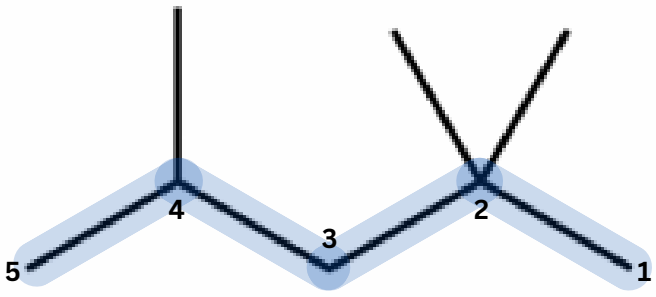
A05

STRUCTURE → NAME

Level: Developing

Molecular Formula: C_8H_{18}

1. NUMBERED PARENT



Parent: 5-carbon chain highlighted · numbered from the end nearer the repeated substituents.

2. NAMING TRAIL

Parent:

Longest chain = 5 C → pentane.

Numbering:

One direction gives locants 2,2,4. The opposite direction gives 2,4,4. First point of difference: 2,2,4 is lower.

Substituents:

Three methyl groups on C2, C2, and C4 → 2,2,4-trimethyl.

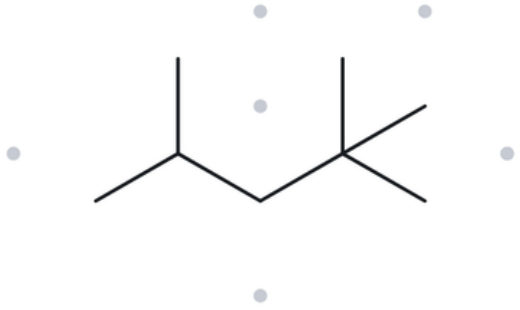
3. MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4. THE ALKANE CHECK

Returned name:

2,2,4-trimethylpentane



✓ Match

Verify

- ☑ longest chain/ring
- ☑ lowest locants
- ☑ aphabetical order
- ☑ punctuation
- ☑ formula counted

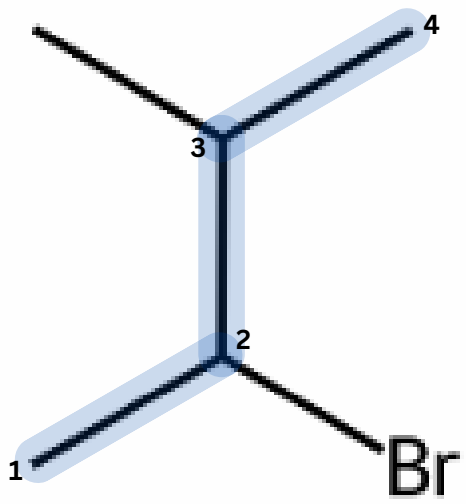
A06

STRUCTURE → NAME

Level: Developing

Molecular Formula: $C_5H_{11}Br$

1. NUMBERED PARENT



Parent: 4-carbon chain highlighted · numbered from the end nearer the bromo substituent.

2. NAMING TRAIL

Parent:

Longest chain = 4 C → butane.

Numbering:

Both directions give locants 2 and 3. The locant sets tie, so use alphabetical order to decide which substituent receives the lower number.

Substituents:

Bromo and methyl. Bromo comes before methyl alphabetically, so bromo receives C2 and methyl receives C3.

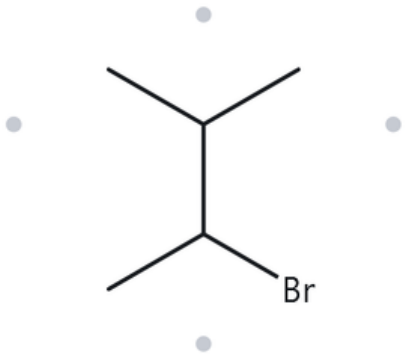
3. MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4. THE ALKANE CHECK

Returned name:

2-bromo-3-methylbutane



✓ Match

Verify ☑ longest chain/ring ☑ aphabetical order ☑ formula counted ☑ lowest locants ☑ punctuation

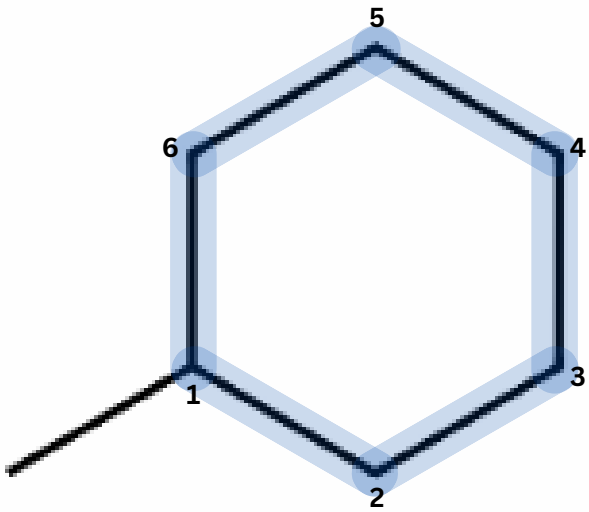
A07

STRUCTURE → NAME

Level: Developing

Molecular Formula: C_7H_{14}

1. NUMBERED PARENT



Parent: 6-carbon ring highlighted · no locant needed for a single substituent.

2. NAMING TRAIL

Parent:

Ring = 6 C → cyclohexane.

Numbering:

Only one substituent is attached to the ring. Its position is automatically C1, so the number is unnecessary.

Substituents:

Methyl attached to the cyclohexane ring.

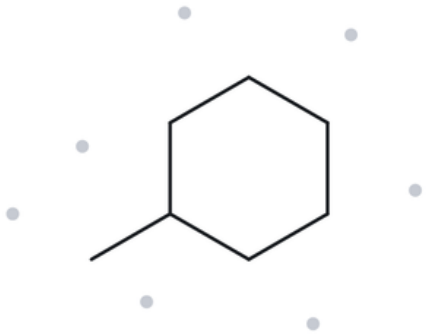
3. MY NAME (WRITTEN FIRST)

methylcyclohexane

4. THE ALKANE CHECK

Returned name:

methylcyclohexane



✓ Match

Verify

- ☑ longest chain/ring
- ☑ lowest locants
- ☑ aphabetical order
- ☑ punctuation
- ☑ formula counted

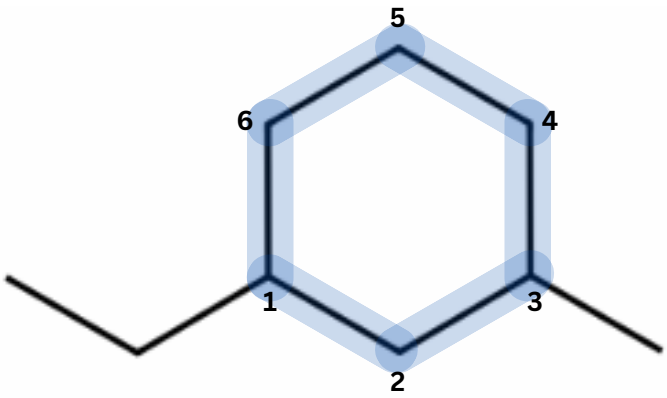
A08

STRUCTURE → NAME

Molecular Formula: C_9H_{18}

Level: Developing

1. NUMBERED PARENT



Parent: 6-carbon ring highlighted · ethyl receives C1.

2. NAMING TRAIL

Parent:

Ring = 6 C → cyclohexane.

Numbering:

Both directions give locants 1 and 3. Since the locant sets tie, alphabetical order decides the numbering. Ethyl comes before methyl, so ethyl receives C1.

Substituents:

Ethyl on C1; methyl on C3.

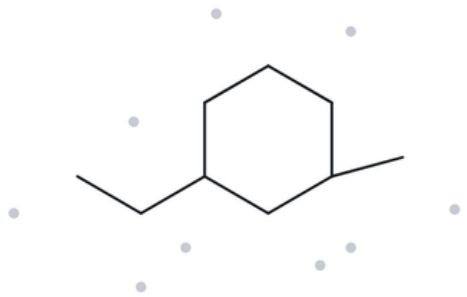
3. MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK

Returned name:

1-ethyl-3-methylcyclohexane



✓ Match

Verify

- ☑ longest chain/ring
- ☑ lowest locants
- ☑ aphabetical order
- ☑ punctuation
- ☑ formula counted

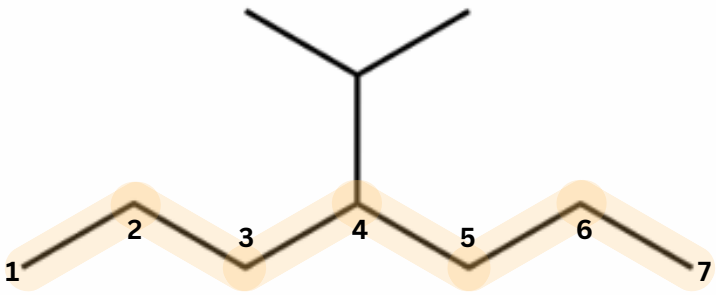
A09

STRUCTURE → NAME

Level: Challange

Molecular Formula: $C_{10}H_{22}$

1. NUMBERED PARENT



Parent: 7-carbon chain highlighted · numbered from either end.

2. NAMING TRAIL

Parent:

Longest continuous chain = 7 C → heptane.

Numbering:

The longest chain runs from the left terminal carbon to the right terminal carbon. Both directions place the branched substituent on C4.

Substituents:

A three-carbon branched group attached through its middle carbon is a (1-methylethyl) substituent on C4.

3. MY NAME (WRITTEN FIRST)

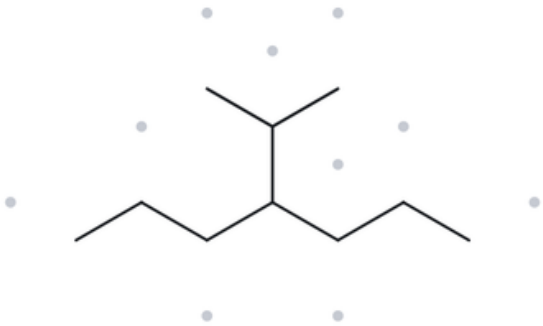
4-(1-methylethyl)heptane

Equivalent substituent name: 4-isopropylheptane.

4. THE ALKANE CHECK

Returned name:

4-(1-methylethyl)heptane



✓ Match

Verify ☑ longest chain/ring ☑ aphabetical order ☑ formula counted ☑ lowest locants ☑ punctuation

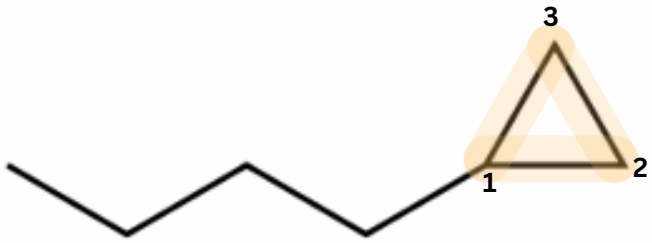
A10

STRUCTURE → NAME

Level: Challange

Molecular Formula: C_7H_{14}

1. NUMBERED PARENT



Parent: 3-carbon ring highlighted · the ring is the parent

2. NAMING TRAIL

Parent:

Ring = 3 C → cyclopropane.

Numbering:

Only one substituent is attached to the ring. Its position is automatically C1, so the number is unnecessary.

Substituents:

A four-carbon straight-chain group is attached to the cyclopropane ring → butyl.

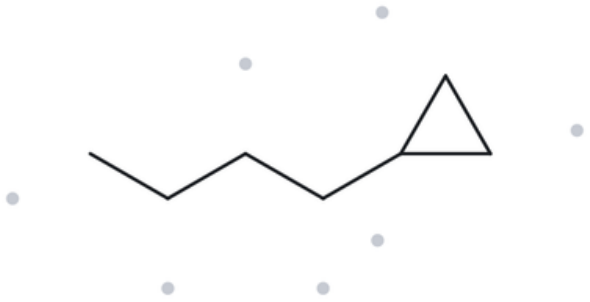
3. MY NAME (WRITTEN FIRST)

butylcyclopropane

4. THE ALKANE CHECK

Returned name:

butylcyclopropane



✓ Match

Verify ☑ longest chain/ring ☑ aphabetical order ☑ formula counted ☑ lowest locants ☑ punctuation

B01

NAME → STRUCTURE

Level: Foundation

Given Formula: 2-methylpropane

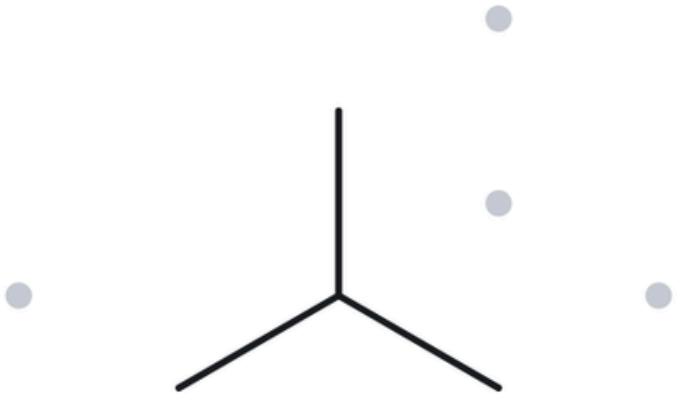
1. NAME DECODED

Parent:
Propane → 3-carbon chain

Substituents:
Methyl on C2

Expected Formula:
3 parent carbons + 1 methyl carbon = 4 C

2. BOND-LINE STRUCTURE (THE ALKANE)



Three-carbon chain with one methyl branch on C2.

3. FORMULA CHECK

Counted From Drawring:
C_4H_{10}

Open Chain:
C_nH_{2n+2}, n = 4 → C_4H_{10}

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

2-methylpropane

✓ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B02

NAME → STRUCTURE

Level: Foundation

Given Formula: 3-ethylpentane

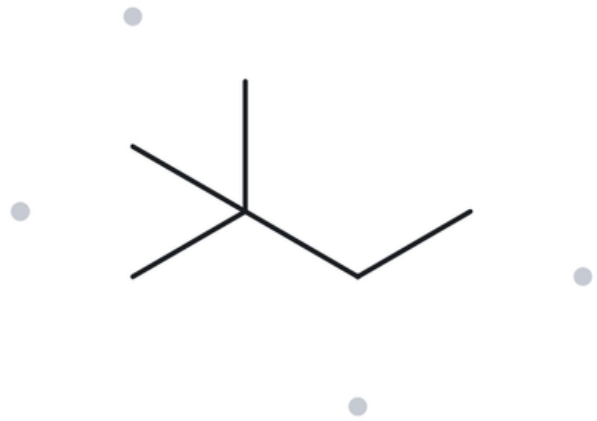
1. NAME DECODED

Parent:
Pentane → 5-carbon chain.

Substituents:
Ethyl on C3.

Expected Formula:
5 parent carbons + 2 ethyl carbons = 7 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Five-carbon chain with a two-carbon ethyl branch on C3.

3. FORMULA CHECK

Counted From Drawring:
C_7H_{16}

Open Chain:
C_nH_{2n+2}, n = 7 → C_7H_{16}

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

3-ethylpentane

✓ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B03

NAME → STRUCTURE

Level: Foundation

Given Formula: 2,2-dimethylbutane

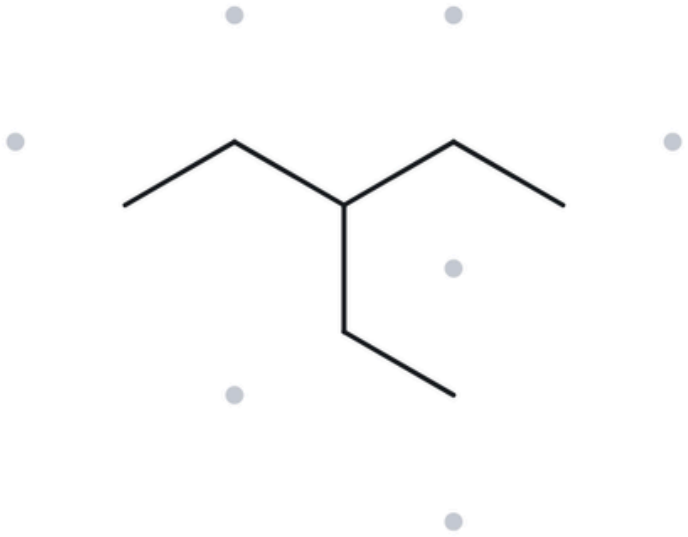
1. NAME DECODED

Parent:
Butane → 4-carbon chain.

Substituents:
Two methyl groups on C2.

Expected Formula:
4 parent carbons + 2 methyl carbons = 6 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Four-carbon chain with two methyl branches attached to C2.

3. FORMULA CHECK

Counted From Drawring:
C_6H_{14}

Open Chain:
C_nH_{2n+2}, n = 6 → C_6H_{14}

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

2,2-dimethylbutane

✓ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B04

NAME → STRUCTURE

Level: Developing

Given Formula: 3-ethyl-2-methylpentane

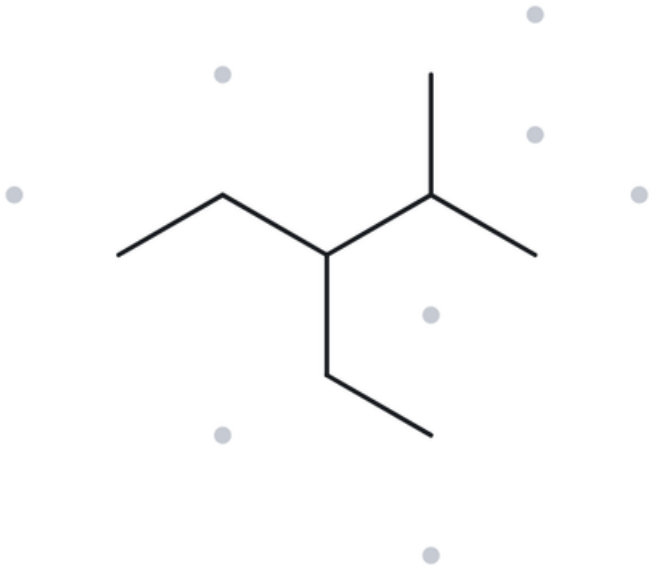
1. NAME DECODED

Parent:
Pentane → 5-carbon chain.

Substituents:
Ethyl on C3; methyl on C2.

Expected Formula:
5 parent carbons + 2 ethyl carbons + 1 methyl carbon = 8 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Five-carbon chain with a methyl branch on C2 and an ethyl branch on C3.

3. FORMULA CHECK

Counted From Drawring:
C_8H_{18}

Open Chain:
C_nH_{2n+2}, n = 8 → C_8H_{18}

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

3-ethyl-2-methylpentane

✓ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B05

NAME → STRUCTURE

Level: Developing

Given Formula: 1-chloro-2-methylpropane

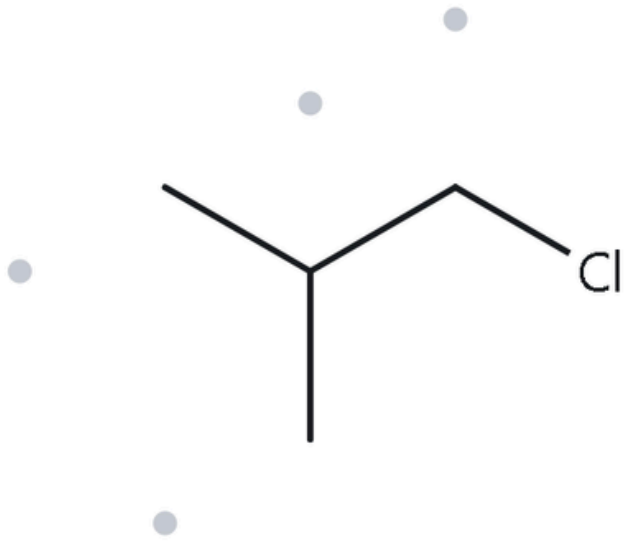
1. NAME DECODED

Parent:
Propane → 3-carbon chain.

Substituents:
Chloro on C1; methyl on C2.

Expected Formula:
3 parent carbons + 1 methyl carbon = 4 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Three-carbon chain with chlorine attached to C1 and a methyl branch on C2.

3. FORMULA CHECK

Counted From Drawring:
C_4H_9Cl

Open Chain:
C_4H_{10}; Cl replaces one H → C_4H_9Cl

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

1-chloro-2-methylpropane

✓ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B06

NAME → STRUCTURE

Level: Developing

Given Formula: 1,2-dimethylcyclopentane

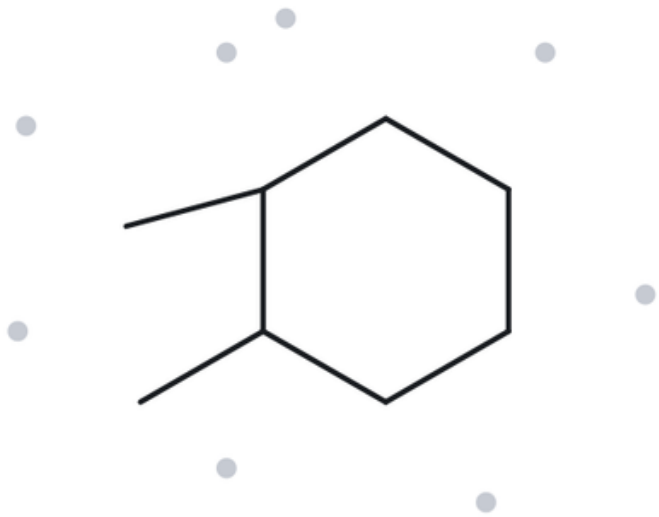
1. NAME DECODED

Parent:
Cyclopentane → 5-carbon ring.

Substituents:
Methyl on C1 and methyl on C2.

Expected Formula:
5 ring carbons + 2 methyl carbons = 7 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Five-carbon ring with methyl branches on two adjacent ring carbons, C1 and C2.

3. FORMULA CHECK

Counted From Drawring:
C_7H_{14}

One ring:
C_nH_{2n}, n = 7 → C_7H_{14}

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

1,2-dimethylcyclopentane

✓ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B07

NAME → STRUCTURE

Level: Developing

Given Formula: 4-ethyl-3,3-dimethylheptane

1. NAME DECODED

Parent:

Heptane → 7-carbon chain.

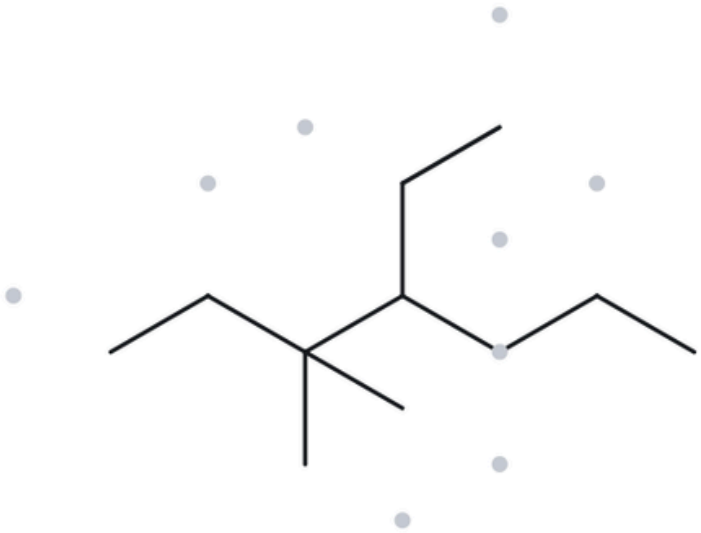
Substituents:

Ethyl on C4; two methyl groups on C3.

Expected Formula:

7 parent carbons + 2 ethyl carbons + 2 methyl carbons = 11 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Seven-carbon chain with an ethyl branch on C4 and two methyl branches on C3.

3. FORMULA CHECK

Counted From Drawring:

$C_{11}H_{24}$

Open Chain:

C_nH_{2n+2}, n = 11 → $C_{11}H_{24}$

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:

4-ethyl-3,3-dimethylheptane

✓ Same as the given name

Verify

☑ parent length matches

☑ substituents position matches

☑ formula agrees

☑ back-check recorder

B08

NAME → STRUCTURE

Level: Challenge

Given Formula: tert-butylcyclohexane

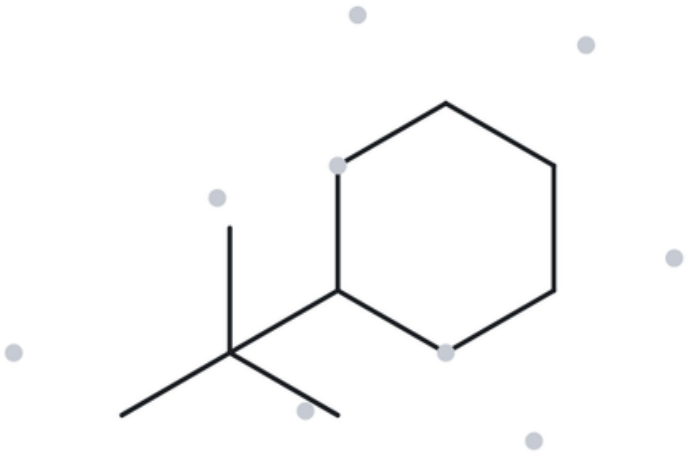
1. NAME DECODED

Parent:
Cyclohexane → 6-carbon ring.

Substituents:
tert-butyl group attached to the ring. The tert-butyl group contains 4 carbons.

Expected Formula:
6 ring carbons + 4 substituent carbons = 10 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Cyclohexane ring with a tert-butyl group attached to one ring carbon. The tert-butyl group has a central carbon attached to three methyl groups.

3. FORMULA CHECK

Counted From Drawring:
$C_{10}H_{20}$

One ring:
C_nH_{2n}, n = 10 → $C_{10}H_{20}$

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:
(1,1-dimethylethyl)cyclohexane

Given name:
(1,1-dimethylethyl)cyclohexane

✗ Same as the given name

Verify

☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B09

NAME → STRUCTURE

Level: Challenge

Given Formula: 3-ethyl-1,1 dimethylcyclohexane

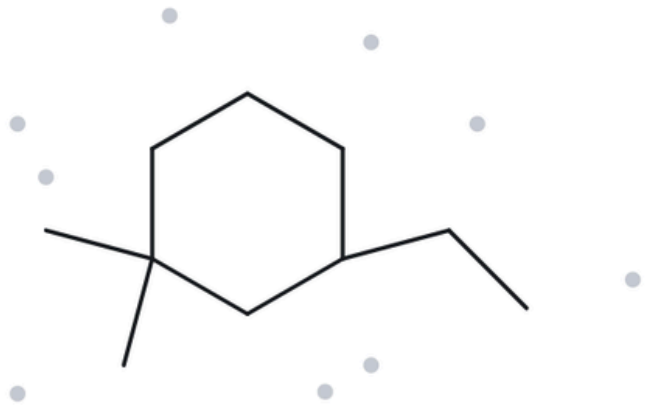
1. NAME DECODED

Parent:
Cyclohexane → 6-carbon ring.

Substituents:
Two methyl groups on C1; ethyl on C3.

Expected Formula:
6 ring carbons + 2 methyl carbons + 2 ethyl carbons = 10 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Cyclohexane ring with two methyl branches on C1 and an ethyl branch on C3.

3. FORMULA CHECK

Counted From Drawring:
$C_{10}H_{20}$

One ring:
C_nH_{2n}, n = 10 → $C_{10}H_{20}$

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:
3-ethyl-1,1 dimethylcyclohexane

✓ Same as the given name

Verify
☑ parent length matches
☑ substituents position matches
☑ formula agrees
☑ back-check recorder

B10

NAME → STRUCTURE

Level: Challenge

Given Formula: 2-ethylbutane (deliberately incorrect name)

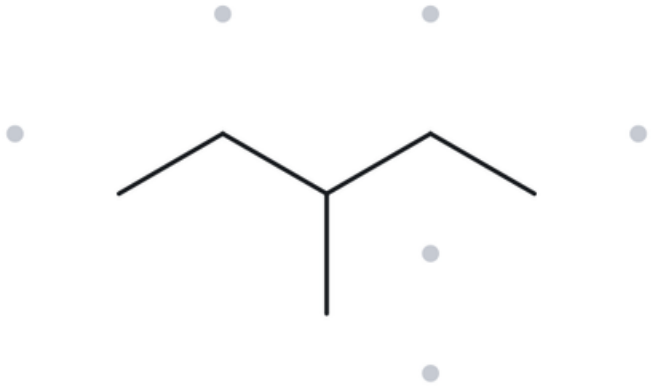
1. NAME DECODED

Parent:
Butane → 4-carbon chain.

Substituents:
Ethyl on C2.

Expected Formula:
4 parent carbons + 2 ethyl carbons = 6 C.

2. BOND-LINE STRUCTURE (THE ALKANE)



Four-carbon chain with a two-carbon ethyl branch attached to C2.

3. FORMULA CHECK

Counted From Drawring:
C_6H_{14}

Open Chain:
C_nH_{2n+2}, n = 6 → C_6H_{14}

☑ agrees with the expected fomula

4. BACK-CHECK

Returned name:
3-methylpentane

Given name:
2-ethylbutane

Rule Broken:
The longest-chain rule. The structure contains a continuous chain of 5 carbons, so the parent should be pentane, not butane.

✗ Same as the given name

Verify

☒ parent length matches
☒ substituents position matches
☒ formula agrees
☑ back-check recorder